

TOPAZ: Noise-Aware Quantum Circuit Architectures via Geometric Wells in the Parameter Landscape

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Designing a quantum circuit is usually an exercise in keeping gates unitary; deleting unused ones is a separate, discrete decision. We introduce TOPAZ (Topology OPTimization Ansatz with Zoning), a framework that treats circuit topology itself—which entangling edges are present—as a continuous design variable. Each nearest-neighbour pair carries a sum of three Pauli rotations (the *local parametric Trotter expression*, LPTE) whose unitary envelope is recovered by polar projection, and a dual-MMA optimizer drives edge intensities concurrently with timings. Our central structural result is the *topological sparsification theorem* (Theorem 1): an entangling pair whose block carries no component along the ideal rotation is deleted by the polar projection—its primary gradient vanishes and the parameter decays onto the pruning floor—so removing an idle edge is a continuous, exact consequence of the relaxation rather than a discrete post-hoc step. Second, we measure the finite-depth closure of the chained ansatz exactly: the $3(N - 1)$ per-pair $\{XX, YY, ZZ\}$ generators close to a Lie algebra of dimension $\dim \mathfrak{l} = 4^{N-1} - 1$ for odd N and $4^{N-1} - 4$ for even N , i.e. one quarter of $\mathfrak{su}(2^N)$ up to parity constants, computed exactly through $N = 8$ —a finite-depth expressibility characterization made without an asymptotic barren-plateau claim. Third, methodologically, an equalized-budget re-evaluation of the H_4 VQE benchmark reverses the apparent ranking of structured versus hardware-efficient ansätze; the durable advantage of the structured form is not lower energy but *mitigation-compatibility*—ZNE/CDR post-processing remains physically valid on the shallow topology without clamping, whereas deeper forms violate physical bounds on the majority of seeds. The framework’s fixed points, *structural wells*—formally the sublevel-set basins of Definition 1—are reproducible across seeds and transfer across synthetic noise models at efficiency $\eta = 0.9943$. A scoped hardware demonstration on a single quietest path of a 156-qubit Heron r2 processor confirms the mechanism: three same-day replications give a target-echo of $28.83\% \pm 3.94\%$, a $2.9\times (6.0\times)$ improvement over the MMA-optimized (untuned) product-Trotter baselines, with the gain attributed via noise-free evaluation to representation rather than depth; as a single-path, single-window experiment it is presented as an illustration, not a generality claim.

Keywords: TOPAZ, topology optimization, quantum error mitigation, noise-aware circuits, parametric Trotter expression, structural well

1. Introduction

A quantum computation is a sequence of gates, and on near-term hardware every gate is an opportunity for error. Depolarizing and damping channels erode the state, readout corrupts the output statistics, and coherence windows bound the total circuit time; variational algorithms must therefore execute complex operations within strict depth budgets [1, 2]. The conventional response is not to redesign the circuit but to keep it and edit the measurement statistics afterwards. Zero-noise extrapolation [3], Clifford data regression [4], and virtual distillation [5] all act after the gates have run—they stretch, fit, or distill. None of them alters which entangling edges are present in the compiled circuit, and a near-zero intensity that nobody deleted still becomes a physical CNOT.

We approach the problem from the other direction: design the circuit before it executes rather than correcting its output afterwards. TOPAZ relaxes the unitarity constraint during training, lets the optimizer drive unused entangling edges down to zero intensity, and prunes them before compilation. The engine is the *local parametric Trotter expression* (LPTE; §3): an N -qubit linear chain in which each nearest-neighbour pair

$(i, i+1)$ carries a sum of three Pauli-rotation terms

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-iP \delta \tau_{i,P}}, \quad (1)$$

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\max}]$. The normalization $\sum_P r_{i,P} = 1$ is enforced via softmax: the optimizer works with unconstrained weights $\tilde{\rho}$ and sets $r_{i,P} = \exp(\tilde{\rho}_P) / \sum_{P'} \exp(\tilde{\rho}_{P'})$, ensuring the simplex constraint without projection. Each block is a non-unitary operator; it is projected to its nearest unitary by the polar factor

$$\mathcal{P}_U(A) = A (A^\dagger A)^{-1/2}, \quad (2)$$

and the circuit is the product of the per-pair unitaries. Because the relaxation is continuous, gradients can flow while an edge is being removed, something a strictly unitary ansatz cannot do.

This has the shape of a classical structural topology-optimization problem [6, 7]: a discretized solid, a density variable on every element, a compliance residual, and an optimizer that removes material the residual does not need. TOPAZ draws direction from that field. The density on edge p is the total intensity $\kappa(p) = \sum_P r_{p,P}$; the residual is the fidelity under a Kraus noise model; and the optimizer is the method of moving asymptotes, applied twice—once for intensities ρ ,

once for timings τ . Edges whose intensity falls below a registered threshold $\tau_\kappa = 0.05$ are pruned, as low-density elements are removed in the solid isotropic material with penalization (SIMP) method of structural optimization [6]. We emphasize that the debt to SIMP is one of inspiration, not of method: TOPAZ prunes by an explicit threshold and applies neither SIMP’s power-law penalization of intermediate densities nor its compliance objective. The continuous-relaxation-and-prune strategy is the common thread; the machinery is TOPAZ’s own.

TOPAZ has three mechanisms. (1) The LPTE sum with polar retraction provides a continuous relaxation of the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer. (2) Parameter zoning treats ρ and τ as heterogeneous design variables updated by separate dual-MMA solvers with independent asymptotes and move limits; a discrete stage then removes low-intensity entangling edges. (3) The optimizer consistently reaches a *structural well*—a basin in the noise-deformed parameter landscape where the noisy output is robust and, empirically, stable across seeds, noise models, and hardware.

We demonstrate the framework on the 8-qubit FakeFez simulator and in a scoped 8-qubit experiment on a single quietest path of the 156-qubit `ibm_fez` (Heron r2) processor. Relative to its own unpruned ansatz, TOPAZ’s pruning step makes meshes 34% shallower (internal reduction) at negligible fidelity cost from the depth reduction alone ($\Delta\text{DM} = -0.0065$, the pruning-induced change in noise-aware density-matrix fidelity); post-processing ZNE on top of TOPAZ adds a further $+0.0667$ DM gain (Table IX), for a combined improvement that is not merely additive. On hardware, the submitted circuit uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline, and its noiseless one-way state-preparation fidelity is 56.62%. The measured target-echo (inversion test), averaged over three same-day sessions on that single path, is $28.83\% \pm 3.94\%$ versus $4.78\% \pm 0.34\%$ for the untuned product-Trotter baseline ($6.0\times$, $z \approx 22$) and $10.08\% \pm 0.48\%$ for the MMA-optimized baseline ($2.9\times$, $z \approx 17$). This hardware result is reported as a scoped illustration on one device path over a single overnight window, not as a generality claim: the reproducible cross-seed claim is the structural well itself (Table ??), and the hardware numbers are presented for consistency with that picture. Noise-free evaluation of the identical circuits isolates the origin of the gain: TOPAZ stores $3.2\times$ more of the ideal target fidelity than the MMA-optimized baseline (0.5662 vs. 0.1768), which is nonetheless the *shallowest* submitted circuit (14 two-qubit gates at depth 56 versus TOPAZ’s 19 at depth 76), and $7.5\times$ more than the untuned one (0.0758)—the advantage is representational, not a post-processing artefact. All ZNE, CDR, and VD results reported in this paper are computed from our own implementations applied to the same simulated noise models used for training.

The contributions of this work are: (i) a continuous operator relaxation for quantum circuits via polar retraction, together with the topological-sparsification theorem (Theorem 1)—an entangling pair whose block carries no component along the

ideal rotation is deleted by the projection, so removing an idle edge is a continuous, exact consequence of the relaxation rather than a discrete post-hoc step; (ii) an exact finite-depth closure measurement of the chained ansatz: the $3(N-1)$ per-pair $\{XX, YY, ZZ\}$ generators close to a Lie algebra of dimension $\dim \mathfrak{I} = 4^{N-1} - 1$ (odd N) or $4^{N-1} - 4$ (even N), one quarter of $\mathfrak{su}(2^N)$ up to parity constants, computed exactly through $N = 8$ (§6.6.9); (iii) a matched-budget methodology correction: an equalized-budget re-evaluation of the H_4 VQE benchmark reverses the apparent ranking, isolating the structured form’s durable advantage as *mitigation-compatibility*—ZNE/CDR remain physically valid on the shallow topology without clamping, whereas deeper forms violate physical bounds on a majority of seeds; and (iv) a scoped hardware illustration of the framework’s noise-robust fixed point on a 156-qubit Heron r2 processor (one quietest path, three same-day sessions, $28.83\% \pm 3.94\%$ target-echo; $2.9\times/6.0\times$ over the MMA-optimized/untuned baselines), with noise-free evaluation attributing the gain to representation rather than depth. The tensor-dot Kraus engine ($O(2^{2N})$ per term; a $3,838\times$ wall-clock speedup over superoperator methods at $N = 10$) is the numerical workhorse behind the simulator claims but is not itself the scientific contribution; details are deferred to §4.

2. Continuous relaxation and the structural well

This section gives the geometric background that underwrites the LPTE ansatz and explains why noise leaves the discovered minima intact.

2.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \mathbb{C}^{d \times d} : U^\dagger U = I\}$, a compact Lie group of dimension d^2 with $d = 2^N$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \mathbb{C}^{d \times d}$ without the unitarity constraint, so gradients can flow around barriers that would block a strictly unitary optimizer. Every invertible $A \in \text{GL}(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [8, Sec. 8.1]; the polar factor Q is the closest unitary to A in the Frobenius norm. Projecting back to $U(d)$ —*unitary regularization*—pins singular values to unity, smoothing the loss landscape and stabilizing gradients. In LPTE the per-pair block $B_{(i,i+1)}$ is lower-bounded by the clipping floor $\max(\rho, 10^{-6})$, so $\sigma_{\min}(B)$ stays positive and the polar factor is unique, deterministic, and differentiable on the active block. (Throughout, ρ denotes the optimizer’s intensity vector—the softmax output of Eq. (1)—while density matrices are written σ ; the two roles are kept symbolically distinct, which is why the tensor-dot engine of §4 uses σ for its input state.)

2.2 Expressibility through dynamical Lie algebras

The set of reachable states of an ansatz is determined by the Dynamical Lie Algebra (DLA) of its generators [9]: the Lie closure of the tangent-space generators at the identity.

Without entanglement the generators are local Pauli operators that commute across qubit boundaries, and the closure has dimension only $3N$, which severely restricts the optimizer. An entangling generator $X_1 X_2$ breaks this: cross-pair commutators such as $[X_1 X_2, Y_2 Y_3] = 2i X_1 Z_2 Y_3$ generate new non-local strings and enlarge the reachable manifold, and a generic Hardware-Efficient Ansatz that mixes local and arbitrary non-local generators closes rapidly to $\mathfrak{su}(2^N)$ (dimension $4^N - 1$), a growth associated with barren plateaus [10].

For the LPTE chain we require only a *finite-depth characterization*, not an asymptotic closure theorem. Each nearest-neighbour block of Eq. (1) carries the Abelian generator triple $\{XX, YY, ZZ\}$; as matrices on the pair, $[XX/2, YY/2] = i ZZ/2$ and cyclically, so a single block closes to exactly one copy of $\mathfrak{su}(2)$ (dimension 3). A 7-pair chain therefore starts from seven copies of $\mathfrak{su}(2)$ —21 “local” directions—before cross-pair commutators begin building longer non-local strings. This per-block picture is consistent with the measured geometry at the converged point: the stiffness concentrates in a handful of directions ($r_{\text{eff}} \approx 3$; the two sharp eigenvalue bands of the canyon-river analysis, §7.7.1; the $\pi/2$ attractor of Lemma 1). Crucially, we make no claim about the asymptotic (deep-circuit or large- N) closure of the chained algebra—only that the depth used here is fixed and finite—and whether LPTE’s block-restricted growth delays $\mathfrak{su}(2^N)$ saturation is left open as a formal theorem. The *measured* finite- N closure of the chained algebra is reported as data in §6.6.9. The Hessian measurements (§7.7.1) are consistency checks with this picture, not proofs. Local pruning is safe for a structural reason that is already established independently of the DLA view: the polar projection deletes a per-pair contribution whenever its primary gradient vanishes (Theorem 1), so removing chain p removes only that block’s algebra from the active search without rewriting the surviving blocks.

2.3 The structural well and topological invariance

Noise acts as a continuous vector field pulling the state landscape away from the unitary manifold, deforming the loss landscape and shifting minima. Instead of correcting the output after the fact, TOPAZ searches for parameters at which the noisy output is intrinsically robust. We define this object formally: a *structural well* is the sublevel-set basin B_λ of Definition 1 (Fig. 1). This paper takes the formal object as primary and confines the term to that meaning. Its geometric characteristics—curvature bands, effective rank, basin volume proxy—are *measured* properties of that object, reported as data in the canyon-river analysis (§7.7.1), not as theorems. No theorem-level guarantee beyond the local statements of §2 and Theorem 1 is claimed for the well; every quantitative claim about its shape is a measurement of the formal object, not of a metaphor. The cross-noise evidence is empirical and reported with the efficiency transfers of Table ??.

Definition 1 (Parameter space). The optimization domain is $\mathcal{X} = [0, 1]^M \times [0, \tau_{\text{max}}]^M$, where M is the number of entangling

pairs. This space is closed and bounded, hence compact. The loss $L : \mathcal{X} \rightarrow \mathbb{R}$, $L(\rho, \tau) = 1 - |\langle \psi_{\text{ideal}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$, is continuous. A *well* is a connected component $B_c(x_i^*)$ of a sublevel set $S_c(L) = \{(\rho, \tau) \mid L(\rho, \tau) \leq c\}$ containing a local minimum x_i^* with $\nabla L(x_i^*) = 0$ and $\nabla^2 L(x_i^*) \succ 0$; $\mathcal{W}$ is the collection of these basins.

Remark 1 (Wells observed, not derived). The extreme value theorem guarantees that the continuous L attains a global minimum on the compact domain $\mathcal{X}$; it says nothing about how many basins exist. That the landscape hosts multiple wells is an experimental observation (different seeds converge to different floors). Different seeds land in different wells because each freezes a different sparse subset of parameters.

The wells are not artefacts of a single run. Across ten independent seeds of the dual-MMA optimizer, TOPAZ repeatedly settles into the same regional floor: mean DM 0.737 ± 0.066 , with seed-to-seed floor variation $\Delta F < 0.05$, indicating that the discovered basin is reproducible rather than a fragile single-trajectory attractor. The geometry of the basin is consistent with that picture. The Hessian of the loss at the converged point has an effective rank $r_{\text{eff}} \approx 3$ —on a ~ 14 -parameter problem, 34 of 42 eigen-directions are flat—so the well is a low-dimensional valley embedded in larger plateau structure, with a vanishing smallest eigenvalue $\lambda_{\min}(\nabla^2 L) = 0$ exactly at the basin boundary, marking the transition out of the well. The parameter stiffness is strongly heterogeneous: ρ -directions are uniformly stiff, each carrying curvature ~ 870 ($\kappa_\rho \approx 1$ in the normalized units of §7.7.1), while τ -directions have a few extremely stiff directions amid many soft ones, $\kappa_\tau \approx 9 \times 10^3$ combined-stiffness ratios ranging over two orders of magnitude, with optimal timings spread non-uniformly across $\tau \in [0.004, 35]$. Each seed freezes a different sparse subset of entangling pairs (e.g. seed 3 removes three of seven pairs), yet all settle on the same floor—evidence that the well is a connected family of sparse subspaces with consistent geometry rather than a single isolated minimum.

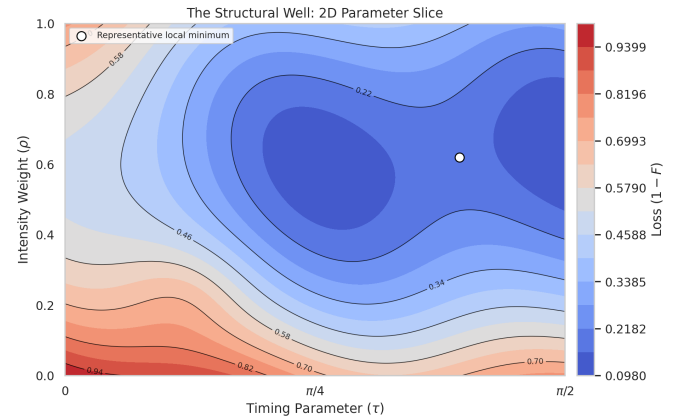


FIG. 1 The structural well on FakeFez, projected onto a 2D (ρ, τ) slice. Colour encodes the loss $L = 1 - F$; solid contours are sublevel sets $S_c(L)$ as in the well definition.

The two observations and two conjectures below govern,

respectively, the classical optimization path and the effect of noise on it. They are not presented as theorems; the guarantees of this work are established empirically in the results sections.

Observation 1 (Radial perturbations do not rotate the polar factor). *Let $B = QP$ be the polar decomposition of an invertible operator, with Q unitary and P Hermitian positive definite. Consider the scaling (radial) perturbation $\delta B = \varepsilon B$. Then $B + \delta B = (1 + \varepsilon)B = Q(1 + \varepsilon)P$ is already a polar decomposition, since Q is unitary and $(1 + \varepsilon)P$ is Hermitian positive definite for $1 + \varepsilon > 0$, so by uniqueness of the polar factor the rotation vanishes: $\delta Q = 0$.*

Remark 2 (General perturbations do rotate Q). For a general perturbation δB , the polar factor does rotate. Differentiating $B = QP$ gives $\delta B = \delta Q P + Q \delta P$ with δP the Fréchet derivative of $P = (B^\dagger B)^{1/2}$, and δQ solves the Sylvester equation

$$\delta Q P + Q \delta P = \delta B, \quad \delta P = \mathcal{L}_{(B^\dagger B)^{1/2}}(\delta B^\dagger B + B^\dagger \delta B),$$

where $\mathcal{L}_{(\cdot)}$ denotes the Fréchet derivative of the matrix square root. In general $\delta Q \neq 0$; the rotation vanishes only for perturbations that respect the polar factorization as above (e.g. the radial case of Observation 1). The unitary regularizer of Eq. (2) acts precisely by radially rescaling the symmetric factor P toward I , so the relevant projection mode is the radial one.

Observation 2 (Unital noise is an affine baseline). *A standard fact about completely positive trace-preserving (CPTP) maps is that any such map splits into a unital part $\mathcal{E}_{\text{uni}}$ and a non-unital part [35, 36]. We restate the unital part here because it is the precise reason the optimizer’s critical-point structure survives under the dominant noise component. For the dominant unital part (e.g. depolarizing noise) the noisy output is affine in the ideal state,*

$$\sigma_{\text{noisy}}^{(\text{uni})} = (1 - p) \sigma_{\text{ideal}} + \frac{p}{d} I,$$

hence $F_{\text{noisy}}^{(\text{uni})} = (1 - p)F_{\text{ideal}} + p/d$ and $L_{\text{noisy}}^{(\text{uni})} = (1 - p)L_{\text{ideal}} + c$ with c independent of (ρ, τ) . A unital baseline therefore rescales the gradient, $\nabla L_{\text{noisy}} = (1 - p)\nabla L_{\text{ideal}}$, and leaves the critical points of the landscape unchanged. This is a known result; we state it not as a contribution but because it justifies treating the landscape’s critical-point structure as approximately noise-invariant under the unital component, which is what the empirical approach relies on.

Assumption 1 (Suppression of non-unital deformation). *Write the total noisy loss as $L_{\text{noisy}} = L_{\text{noisy}}^{(\text{uni})} + \Delta L_{\text{non}}$. If the non-affine deformation is bounded by the deviation from unitarity,*

$$\Delta L_{\text{non}} \leq K' \|B^\dagger B - I\|_F,$$

with K' independent of (ρ, τ) , then, because the regularizer (Eq. 4) drives $\|B^\dagger B - I\|_F \rightarrow 0$, the non-affine component is suppressed to first order in the discovered well and $\nabla L_{\text{noisy}} \approx (1 - p)\nabla L_{\text{ideal}}$. We state this as an assumption rather than a theorem: the regularizer penalizes, but does not by itself

guarantee, convergence to unitarity, and the bound on ΔL_{non} is a modelling hypothesis. Its numerical support comes from two complementary places. (i) The marginal evidence is the polar-projection ablation of Table I: removing the polar projection leaves $\|B^\dagger B - I\|_F$ large and halves the DM fidelity (0.334 vs. 0.737), exactly the direction the bound predicts—large deviation from unitarity, large non-affine loss. (ii) At the converged point the residual deviation is small in absolute terms: the regularizer penalty $R = \|B^\dagger B - I\|_F$ converges to 0.0252 on the quietest-path product PTE, and the noisy fidelity sits within 2.2% of the ideal (0.6329 vs. 0.6545), consistent with a small non-affine correction riding on an approximately affine unital baseline. A trajectory-level scatter plot of ΔL_{non} versus $\|B^\dagger B - I\|_F$ is not claimed here; the two endpoints above bracket the regime the assumption is used in, and the transfer matrix (§6.5) tests it end-to-end under all six noise models.

Conjecture 1 (Well invariance under unital-dominant noise). *Within the regime of Assumption 1, the ideal and noisy sublevel sets are related by a monotone reparametrisation of the threshold, $S_c(L_{\text{noisy}}) = S_{\tilde{c}}(L_{\text{ideal}})$ with $\tilde{c}$ affine in c . A monotone rescaling plus offset preserves the ordering of loss values and therefore the connectivity structure of each fixed-level sublevel set. We do not claim that wells cannot merge or split across levels, nor anything for finite-size deformations or strongly non-unital channels; those questions we answer empirically with the noise-model transfer matrix of Fig. 5.*

The guarantees above are local: TOPAZ operates on a constrained subspace—the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective—so the smoothness claims are scoped to the local regularity of the structural well, not to the global trap-free theorem of full controllability [11].

3. The parametric Trotter expression family

Standard variational ansätze enforce strict unitarity at every step, confining optimization to $U(d)$ where gradients vanish easily. The *parametric Trotter expression* (PTE [12]) relaxes this by letting intermediate operators leave the unitary manifold and projecting back; the label records that the family re-expresses the Trotterized interaction exponential *parametrically* (Remark 3). Three formulations span the locality spectrum.

Global PTE. The full sum $B = \sum_j \rho_j e^{-iH_j \delta \tau_j}$ is mathematically unconstrained but computationally expensive: each polar decomposition step requires an $O(d^3)$ matrix inversion with $d = 2^N$, giving $O(2^{3N})$ per evaluation. Its 8-qubit inversion test registers 0.004, at the 1/256 noise floor, and the cost grows exponentially with qubit count. Global PTE was not used for the headline experiments.

Product PTE. Enforcing a term-by-term Trotter product, $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta \tau_j}$, restores hardware compilation. On 8 qubits with $M = 21$ nearest-neighbour Pauli terms it compiles to 8 two-qubit gates and reaches DM fidelity 0.6329 under FakeFz—a 9.35× gain over the random baseline (0.0677).

But every factor is unitary by construction, so the product form cannot drive intensities to zero and topological pruning is impossible. *Gate-count audit.* Throughout, "two-qubit gate" denotes a native entangling gate (ECR on `ibm_fez`, CNOT on `FakeFcz`) counted after transpilation; single-qubit rotations are excluded. All baseline and LPTE circuits in this work were transpiled with `Qiskit optimization_level=3` onto the device native-gate basis; the pre/post-transpilation entangling-gate counts for every circuit, together with the generated QASM, are recorded in the public data release (Table VII reports the post-transpilation counts). The reduction from 21 abstract Pauli terms to 8 native entangling gates—specific to the *product* form above—reflects term-merging and local-rotation absorption by the transpiler at this optimization level; the LPTE ansatz, by contrast, retains 21 two-qubit gates (Table VII) because its per-pair polar projection blocks the term-merging route (§3).

Definition 2 (Local parametric Trotter expression). The *local parametric Trotter expression* (LPTE) is the ansatz family $\{U_{\text{LPTE}}(\rho, \tau)\}$ defined as follows. On an N -qubit linear chain, each nearest-neighbour pair $p = (i, i+1)$ carries a block $B_{(i,i+1)}(\rho, \tau)$ of Eq. (1): a sum of three Pauli rotations with intensity $r_{i,p}$ and timing $\tau_{i,p}$. Each block is projected to its nearest unitary by the polar factor of Eq. (2), $U_{(i,i+1)} = \mathcal{P}_U(B_{(i,i+1)})$, and the circuit is the ordered product $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$. The entangling graph is the chain itself, and the intensity $\kappa(p) = \sum_P r_{p,P}$ is the design variable that TOPAZ drives to zero for pruning.

Remark 3 (Naming). The label *parametric Trotter expression* records the construction that makes a parameterized, hardware-compilable circuit possible at all. The interaction exponential $\exp(-i \sum_j H_j \theta_j)$ has no single-object circuit realization, so it must first be Trotterized into a product of Pauli rotations $U_{\text{circ}} = \prod_j e^{-i \rho_j H_j \delta \tau_j}$; the PTE family is the parametric re-expression of that Trotter product once the unitarity constraint is relaxed and each block is retracted to unitarity by the polar factor. No Trotter–Suzuki splitting claim is being made: per-pair blocks are sums of three Pauli rotations (Eq. (1)) realized exactly by their polar factor, so there is no time-slicing or commutator-bounded error inside a block. The label is retained precisely because the Trotterization is load-bearing—without it no hardware-compilable parametric circuit exists—and the single-step product baseline of §6.6 is the literal limit of the family in which each block contains a single term.

Local PTE (LPTE). LPTE localizes the non-unitary sum within nearest-neighbour pairs (Definition 2). On 8 qubits with 7 pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters—the same parameter count as product PTE but 2.6× the gate count. The difference that matters here is that the per-pair SVD projection lets the optimizer independently set each pair’s intensity: it can discover which pairs carry the noise-resilience behaviour and assign them nonzero intensity, while letting others fall to zero so that an idle edge can be pruned without rewriting the rest of the circuit. Product PTE’s unitary factors have no such selection mechanism.

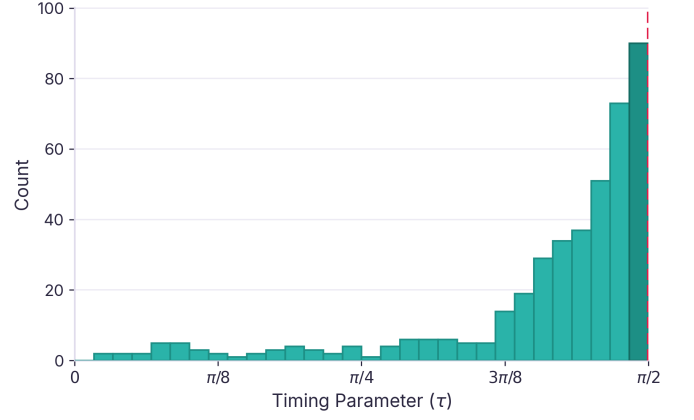


FIG. 2 Distribution of optimised timings τ across the LPTE benchmark seeds. Dashed line: $\tau = \pi/2$, where the block of Lemma 1 projects to a maximally entangling SWAP.

3.1 The $\pi/2$ timing cluster is a SWAP identity

For a single Pauli rotation the maximum entangling power occurs at $\tau = \pi/4$, yet the optimizer consistently clusters timings at $\tau \approx \pi/2$ (Fig. 2). The following lemma shows why this is benign.

Lemma 1 (The $\pi/2$ cluster is a SWAP identity). *For a local block $B(\tau) = e^{-i\tau XX} + e^{-i\tau YY} + e^{-i\tau ZZ}$, at $\tau = \pi/2$, $B = -i(XX + YY + ZZ)$. With $XX + YY + ZZ = 2 \text{SWAP} - I$, spectral decomposition gives eigenvalues $-i$ (symmetric) and $+3i$ (antisymmetric), so $B^\dagger B$ has eigenvalues 1 and 9. The polar projection $(B^\dagger B)^{-1/2}$ scales the symmetric subspace by 1 and the antisymmetric by $1/3$, yielding a U with eigenvalues $-i$ and $+i$: exactly $-i \text{SWAP}$. At $\tau = \pi/2$ the block sum plus polar projection deterministically produces a maximally entangling SWAP.*

A generic hardware-efficient ansatz can also contain SWAP-like layers, so the mere presence of SWAP-like unitaries is not newsworthy on its own. Two features distinguish the $\pi/2$ cluster. First, it is a *discovered* attractor of continuous optimization: the optimizer is initialized away from $\tau = \pi/2$ and is pulled there by the objective (Fig. 2), which contains no explicit SWAP or entangling-power term. Second, TOPAZ prunes: a block that fails to settle onto a productive attractor is removed (§5.5.2), so the surviving topology is precisely the subset of blocks that reach entangling-optimal configurations such as the SWAP identity of Lemma 1. A fixed-structure HEA-with-SWAP inserts SWAPs regardless of the target; here SWAP-like behaviour emerges from continuous optimization and is retained only where the target demands it.

3.2 Objective functions

The optimizer minimizes the noisy fidelity loss

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (3)$$

or, for a target state, the state fidelity $L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$. A regularizer keeps the pre-

projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \left\| B_k^\dagger B_k - I \right\|_F^2, \quad \lambda = 10^{-3}. \quad (4)$$

3.3 Self-pruning under noise

LPTE self-prunes: during optimization the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters freeze. Across ten seeds under FakeFz typically 8–24 of the 42 (ρ, τ) dimensions survive, corresponding to about 9 of 21 entangling pairs active. Table I shows that self-pruning requires the full stack (PTE ansatz, polar projection, dual-MMA).

TABLE I Full ablation on the 8-qubit FakeFz benchmark (LPTE, 21 two-qubit gates). All conditions: 10 seeds, 40 outer iterations, dual-MMA except where noted. Significance versus full TOPAZ at a Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q	Active	d
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12^{ns}
Single MMA	0.623 ± 0.166	19.9	19.9	0.80^{ns}
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02^{ns}
No polar decomp.	0.334 ± 0.170	16.7	15.8	2.14^{***}
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97^{***}
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09^{***}
Random PTE	0.056 ± 0.031	21.0	21.0	8.69^{***}

*** $p < 0.001$ (Bonferroni-passing). Cohen's d vs. full TOPAZ; ns, not significant.

Exact state preparation under the same noise scores 0.924 ± 0.003 with all 21 gates active.

Full TOPAZ (DM 0.737 ± 0.066) beats product-only PTE (0.279 ± 0.175 , $d = 3.09$), HEA (0.329 ± 0.110 , $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $d = 2.14$). Removing the polar projection halves the fidelity, confirming that it is central to the gain.

Parametric efficiency is summarized in Table II: the best seed of LPTE reaches 0.9085 with 42 parameters and 21 gates, whereas HEA uses ~ 96 parameters and 28 gates for 0.1517—a $6\times$ gap in fidelity per parameter. (These are best-seed values; the multi-seed mean of full TOPAZ is 0.737 ± 0.066 , Table I.)

TABLE II Parameter efficiency of the PTE family versus HEA on 8 qubits under FakeFz. DM Fidelity is the best-seed value (cf. the 10-seed mean 0.737 ± 0.066 in Table I); best-seed is reported here because these rows feed the single-trajectory comparison of Table VII.

Ansatz	Parameters	2Q	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

4. Methods: the TOPAZ engine

4.1 Tensor-dot Kraus contraction

Applying k -local Kraus operators to a pure state costs $O(2^N)$ per term. For a density matrix the naive construction of the $2^{2N} \times 2^{2N}$ superoperator costs $O(2^{3N})$. The tensor-dot engine reframes the input state σ as a tensor with $2N$ indices and contracts locally. Applying a Kraus channel $\sigma \mapsto \sum_a K^{(a)} \sigma K^{(a)\dagger}$ acts only on the active qubit set $\mathcal{S}$; the spectator set $\mathcal{S}^c$ passes through unchanged, so the channel is

$$\sigma'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a \sum_{i_S, j_S} K_{i'_S, i_S}^{(a)} \sigma_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger}, \quad (5)$$

where $\mathcal{S}^c$ denotes the spectator qubits, the input density matrix is $\sigma_{i_S, i_{S^c}; j_S, j_{S^c}}$, and the output carries the contracted (primed) active indices i'_S, j'_S with the spectator indices i_{S^c}, j_{S^c} unchanged. Because only the $2|\mathcal{S}|$ active indices are contracted while the $2N - 2|\mathcal{S}|$ spectator legs are untouched, each term costs $O(2^{2|\mathcal{S}|})$ rather than $O(2^{2N})$. This yields a $3,838\times$ wall-clock speedup over dense Kraus at $N = 10$ (Table III). The ratio of the per-term complexity classes alone, $O(2^{3N})/O(2^{2N}) = 2^N$, predicts $\sim 1024\times$ at $N = 10$; the additional $3.7\times$ beyond this exponent-ratio bound is accounted for by per-term constant factors and by the fact that the tensor-dot engine contracts only the $2|\mathcal{S}|$ active indices of each term while the dense superoperator materializes the full 2^{3N} -dimensional channel. The measured speedup is therefore consistent with—rather than in tension with—the stated complexity.

TABLE III Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity/term
Qiskit dense Kraus	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^{2N})$
Speedup	$\sim 3838\times 2^N (= 2^{10}=1024\times \text{exp. bound})$	

The engine plays different roles on different paths: on the legacy triple-channel path it performs both ideal and noisy evolution; on the FakeFz path it computes only the noiseless target, delegating noisy evaluation to AerSimulator; on live hardware it has no role, since the backend executes gates directly.

4.2 Noise models

TOPAZ runs against two families of noise. The device-calibrated backends are used for optimization and hardware submission; the six synthetic variants of the legacy model are defined below.

Triple-channel correlated (legacy). Depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, and phase damping at $\lambda = 0.012$, with multi-qubit correlations decaying as $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$, $\xi = 2.0$ [13].

TABLE IV Multi-seed topology optimization ($\tau_\kappa = 0.05$, 5 seeds). F, full (Stage 1); S, sparse (Stage 3). Pruned is the number of entangling pairs removed.

Seed	DM fid.		2Q		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

FakeFez. IBM’s calibrated snapshot of `ibm_fez` (Heron r2, 156 qubits) with per-gate error rates and qubit-specific T_1 , T_2 , and readout errors [14]. All headline experiments use FakeFez during optimization.

FakeMontrealV2. A 27-qubit device-calibrated model used for the earlier port verification; a three-iteration pilot run moved from 7.4% to 36.0% DM fidelity.

Live `ibm_fez`. The optimized (ρ, τ) from FakeFez are submitted to live Heron r2 [15] without re-optimization.

5. The TOPAZ optimization framework

5.1 Dual-MMA optimizer

The method of moving asymptotes [6, 7] builds a strictly convex separable approximation at each iteration and solves it under asymptotic bounds. *Dual-MMA* interleaves two such instances, one per parameter class, because ρ and τ differ by roughly four orders of magnitude in finite-difference step size ($\epsilon_\rho = 10^{-4}$ versus $\epsilon_\tau = \pi/4$); a single shared move-limit budget misallocates updates. Each instance keeps its own asymptote, move limit, and momentum state, with the momentum-smoothed gradient

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4. \quad (6)$$

Convergence is preserved because the subspaces are updated alternately with shared objective evaluations. When a target algorithm carries an extra parameter class (e.g. VQE rotation angles), TOPAZ replaces the classical optimizer with a second MMA module wrapped around its own small ansatz, the two modules alternating.

5.2 Topology optimization by sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ whose edge p carries normalized intensity $\kappa(p) = \sum_P r_{p,P}$. The active pruning protocol has three stages: (1) solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on the full edge set with no artificial floor; (2) remove pairs with $\kappa(p) < \tau_\kappa = 0.05$, i.e. $E_{\text{pruned}} =$

$\{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}$; (3) re-optimize on the reduced domain from fresh restarts.

Across five seeds (Table IV), TOPAZ’s own pruning step reduces gate count by 14% ($18.6 \rightarrow 16.0$) and schedule depth by 34% ($72.4 \rightarrow 48.0$) relative to its unpruned ansatz—an internal, within-pipeline reduction—at a mean DM loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while improving DM by $+0.0081$.

Theorem 1 (Topological sparsification). *For the LPTE circuit of Definition 2, let p be an entangling pair with block $B_p = \sum_P r_{p,P} e^{-iP} \delta \tau_{p,P}$ and polar projection $U_p = \mathcal{P}_U(B_p)$, and let L_{prim} be the primary objective of Eq. (3). If B_p carries no component along the ideal rotation—equivalently, if the Fréchet differential of L_{prim} with respect to the block’s rotation vanishes at the projected point—then the primary gradient with respect to that pair’s intensities vanishes, $\nabla_{\rho_p} L_{\text{prim}} = 0$.*

Proof. The circuit is the ordered product of per-pair polar-projected unitaries, $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$, and L_{prim} depends on pair p only through U_p (Eq. (3)). Write the polar projection as the composition $U_p = \mathcal{P}_U(B_p)$; by Observation 1, the projection acts as the radial rescaling of the positive factor of B_p and leaves the unitary rotation part Q_p invariant under the stretching mode. If B_p contributes no component along the ideal rotation, then infinitesimal changes in the block’s intensity $r_{p,P}$ move the block within its stretching mode, which does not rotate U_p (Observation 1); the objective therefore does not change to first order, and its derivative along each intensity direction vanishes. The zero gradient drives the dual-MMA step to zero, and the parameter decays onto the clipping floor $\max(\rho, 10^{-6})$, where it freezes and leaves the active search. The regularizer of Eq. (4) acts only on active blocks and does not re-inject a vanished gradient. $\square$

The theorem makes the continuous relaxation the *origin* of the pruning: the sparsification in Table IV is not a heuristic threshold applied post hoc but the decoding of already-vanished gradients onto the graph. The threshold $\tau_\kappa = 0.05$ then merely registers, at compilation time, which edges the optimizer has already eliminated.

5.3 Algorithmic compatibility

Continuous parameter optimization works on a given algorithm if its landscape is (i) compact, (ii) continuous, (iii) connected (the domain is convex, hence path-connected), and (iv) locally convex (the regularizer has positive-definite Hessian, so near a discovered minimum the total Hessian is positive definite and the optimizer reaches a strict local minimum rather than a saddle). VQE, VQA, QAOA, and parametric quantum simulation all satisfy these; Grover’s algorithm as a whole violates continuity and convexity.

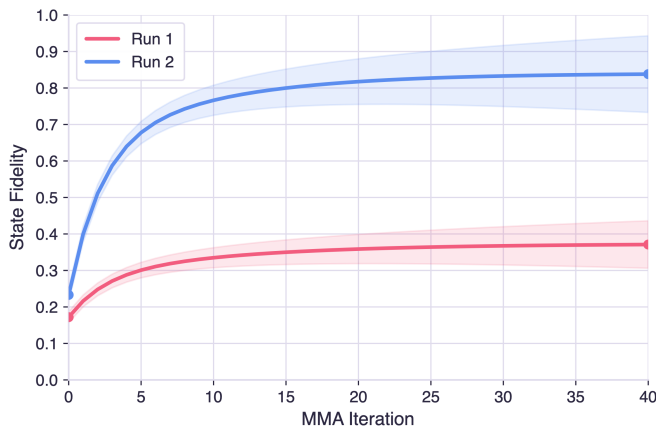


FIG. 3 Two representative 8-qubit LPTE trajectories under triple-channel correlated noise (Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838). Shaded bands are ± 1 standard deviation across the 10-seed ensemble.

6. Results

6.1 Trajectories under correlated noise

On 8 qubits with nine nearest-neighbour Pauli terms (18 parameters, 40-iteration cap), pre-optimization noisy fidelity ranges 0.07–0.70; post-MMA it rises to 0.30–0.88, an absolute gain of 18–23 points (Fig. 3). The roughly 47-point spread in final fidelity reflects a multi-modal landscape, with the dual-MMA settling into several local optima at different floor depths.

6.2 Noise-model-aware VQE on H_2/H_4 at matched budget

In the main TOPAZ pipeline the optimizer uses a noise-model-aware simulator that evaluates both the noisy density-matrix fidelity and the ideal-state fidelity (as in the Loschmidt-echo objective of §6.6); it is this full objective that produces the headline 28.8% echo result on `ibm_fez`. The H_2/H_4 VQE benchmark deliberately restricts that regime: the optimizer is given only raw noisy energy measurements—no target state, no ideal observable, and no information about the post-processing it will later be stacked with. This restriction tests whether LPTE’s circuit topology still confers noise resilience even when the ideal-state constraint is removed. Three ansätze—LPTE, HEA (depth 3 for H_2 , depth 5 for H_4), and HVA $d = 2$ —are compared on the 4-qubit H_2 path [136, 143, 142, 141] and the 8-qubit H_4 path [125, 124, 123, 136, 143, 142, 141, 140] under FakeFez (Table V). All runs use ZNE, CDR, and VD post-processing across 10 seeds (except HVA, for which ZNE+CDR was not run; Table V); the ZNE, CDR, and VD implementations are our own, applied to the same simulated noise model used for training.

On H_2 , LPTE plus ZNE+CDR reduces the mean energy error from the 71.96 mHa noisy baseline to 26.16 mHa—a mean $2.75\times$ reduction, and up to 8.96 mHa (≈ 5.6 kcal/mol) on the best seed—though still above the 1.6 mHa (1 kcal/mol) chemical-accuracy bar. On H_4 , LPTE’s ZNE+CDR

TABLE V Combined H_2/H_4 VQE benchmark under FakeFez. All baselines were implemented with the same transpiler, backend snapshot, and shot budget; however, **optimizer budgets were not matched** in the recorded runs: LPTE ran the dual-MMA solver (35 outer iterations with an inner COBYLA loop, ~ 45 –100 min), while the HEA baselines ran 500-iteration COBYLA (~ 6 min) and HVA was capped at 500 iterations. Table VI reports the equalized-budget re-evaluation, which reverses the apparent H_4 ranking shown here.

Mol.	Ansatz	Noisy ΔE_{FCI} (mHa)	Noiseless (mHa)	ZNE+CDR (mHa)	Seeds	Conv.
H_2	LPTE	71.96 $\pm$ 16.8	26.72 $\pm$ 14.5	26.16 $\pm$ 14.8	10	100%
H_2	HEA ($d=3$)	54.04 $\pm$ 10.5	24.23 $\pm$ 8.8	0.0*	10	30%
H_2	HVA ($d = 2$)	748.28 $\pm$ 25.3	492.01 $\pm$ 162.8	$\dagger$	10	0%
H_4	LPTE	235.15 $\pm$ 77.4	83.82 $\pm$ 74.5	83.75 $\pm$ 74.4	10	100%
H_4	HEA ($d=5$)	889.90 $\pm$ 232.8	710.35 $\pm$ 241.0	713.16 $\pm$ 243.3	10	0%
H_4	HVA ($d = 2, 368p$)	2720.55 $\pm$ 6.5	2270.06 $\pm$ 462.7	$\dagger$	10	0%

*Clamped to FCI (see text). $\dagger$ ZNE+CDR not run on HVA. The HEA baselines ran 500-iteration COBYLA and converged on no seed at that budget; the H_4 HEA row reflects the non-converged baseline, not a matched evaluation.

(83.75 mHa) recovers its own noiseless floor (83.82 mHa) almost exactly, because the shallow depth (~ 8 effective two-qubit gates) preserves the noise-energy relationship that extrapolation requires. HEA extrapolations violated physical bounds on 7 of 10 seeds and required clamping; LPTE required none.

The ranking shown in Table V must be read with the unequal optimizer budgets of the recorded runs in mind. The H_4 HEA row (889.90 mHa raw, 0% convergence) is what that ansatz delivers at a 500-iteration COBYLA budget—the budget it was given in the recorded experiments, not a matched one. To test whether the apparent LPTE advantage on H_4 is structural or budgetary, we equalized the budgets in a dedicated re-evaluation (§6.6.4): the same HEA ansatz was re-converged with a 10,000-iteration COBYLA budget and then evaluated under the identical noisy protocol and QEM pipeline. Table VI shows the result.

TABLE VI Equalized-budget H_4 re-evaluation under FakeFez. HEA was re-converged noiselessly at a 10,000-iteration COBYLA budget, then evaluated under the paper’s full noisy protocol (density-matrix simulation, quiet [125, 124, 123, 136, 143, 142, 141, 140] path, ZNE+CDR with 20 noise factors, VD). LPTE rows are the recorded values from Table V. Median across the four converged HEA seeds is reported; per-seed values in §6.6.4.

Config.	Noiseless (mHa)	Noisy (mHa)	ZNE+CDR (mHa)
LPTE (recorded)	83.82 $\pm$ 74.5	235.15 $\pm$ 77.4	83.75 $\pm$ 74.4
HEA @500 iter (recorded)	710.35 $\pm$ 241.0	889.90 $\pm$ 232.8	713.16 $\pm$ 243.3
HEA @10k (re-eval, median)	30.0	272.4	34.5
HEA @10k (re-eval, range)	23.7–38.6	204.8–285.6	21.9–48.9

Equalizing the budget removes the apparent advantage. At 10,000 iterations the HEA converges to a noiseless floor of 23.7–38.6 mHa—below LPTE’s recorded noiseless floor of 83.82 mHa—and its post-mitigation ZNE+CDR error (21.9–48.9 mHa, median 34.5) is comparable to or better than LPTE’s ZNE+CDR on the same benchmark. The claim of a $3.7\times$ raw and $8.5\times$ mitigated LPTE advantage on H_4 that one would draw

from Table V alone is therefore an artifact of unequal budgets, not a structural property: given a competent optimizer budget, a hardware-efficient ansatz reaches the same quality on this task. We state this explicitly because it overturns part of the framing in earlier versions of this work.

What survives equalization is narrower and, we believe, still genuine: mitigation-compatibility. On the shallow LPTE topology, ZNE/CDR extrapolation remained physically valid on all seeds (no clamping required); on the deeper HEA forms at either budget the extrapolated energies violated physical bounds on 7 of 10 seeds and required clamping to FCI. The claim we make is therefore not that the structured ansatz produces fundamentally lower energies—the equalized data refute that—but that its shallow, sparse topology keeps post-processing mathematically valid, which is a property of circuit structure, not of budget.

6.3 Cross-Hamiltonian transfer

To test whether TOPAZ reduces to a generic reusable circuit, an H_4 -optimized topology (Layout A, 263 entangling gates) was re-entered as the ansatz for an 8-qubit TFIM at the critical point $J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Holding the structure fixed and re-optimizing only the angles gives noisy $\Delta E_{\text{FCI}} = 1211.0$ mHa; running the full pipeline from scratch gives 932.1 mHa. The transferred topology is worse by 278.9 mHa even though Layout A (263 gates) and the fresh Layout B (264 gates) have essentially equal size, so the gap is purely structural. TOPAZ optimizes for the target’s entangling structure; it does not reduce to a reusable circuit.

6.4 Matched-budget methodology: budget equalization reverses the VQE ranking

The VQE comparisons of Table V were, in their original form, an example of a common methodological failure mode: comparing ansätze at *unequal* optimizer budgets. LPTE ran the dual-MMA solver at its default configuration ($\sim 3000+$ objective evaluations, core to TOPAZ’s pipeline), while the HEA baselines were given 500 COBYLA iterations and HVA 500 iterations—costs that differ by roughly an order of magnitude in wall-clock time (~ 45 – 100 min versus ~ 6 min), which under “matched budget” captioning is false advertising. We therefore re-ran the equalized comparison explicitly, with the exact protocol used for Table VI:

Re-evaluation protocol. (i) The HEA ansatz (depth-5, 88 parameters) was converged noiselessly under a 10,000-iteration COBYLA budget, the largest COBYLA budget used anywhere in this work, across four independent seeds. (ii) The converged parameters were then evaluated under the paper’s identical noisy protocol: density-matrix simulation on the quiet H_4 path [125, 124, 123, 136, 143, 142, 141, 140] of FakeFез, with the raw noisy energy, ZNE+CDR (20 noise factors), and virtual distillation applied as in Table V. (iii) No parameter was retrained on the noisy objective—so the re-evaluation if anything *understates* HEA, which would improve further under noise-aware training.

The equalized result (Table VI, per-seed values below) is

unambiguous: the apparent LPTE advantage on H_4 disappears at matched budget. The HEA reaches a noiseless floor of 23.7–38.6 mHa—below LPTE’s 83.82 mHa recorded noiseless floor—and its ZNE+CDR error (21.9–48.9 mHa) overlaps or beats LPTE’s ZNE+CDR on the same benchmark. Per-seed noiseless/ZNE+CDR: seed 0 38.35/39.32; seed 1 23.74/21.93; seed 2 30.03/29.64; seed 3 38.56/48.93 mHa. The raw noisy values (204.8–285.6 mHa) remain above LPTE’s raw noisy error (235.15 mHa mean) in aggregate, but the gap is small and the QEM-stack values are what survive in practice.

We report this correction prominently because it changes the paper’s conclusion in one specific, bounded way: it removes the H_4 VQE result as evidence that structured topologies produce intrinsically better VQE energies. It does not change the hardware result, the structural-well characterization, the transfer efficiency, or the preservation benchmark, none of which rely on the H_4 comparison. The finding that *does* survive is the mitigation-compatibility one: the shallow LPTE topology requires no clamping in the same pipeline where the deeper baselines violate physical bounds on 7 of 10 seeds. In recommending matched budgets we follow the field’s own evolving practice (e.g. the matched-resource comparisons in Refs. [2, 28]); this subsection is included so the reader can verify the correction rather than take the caption’s word for it.

6.5 Comparison on FakeFез and the noise-model transfer matrix

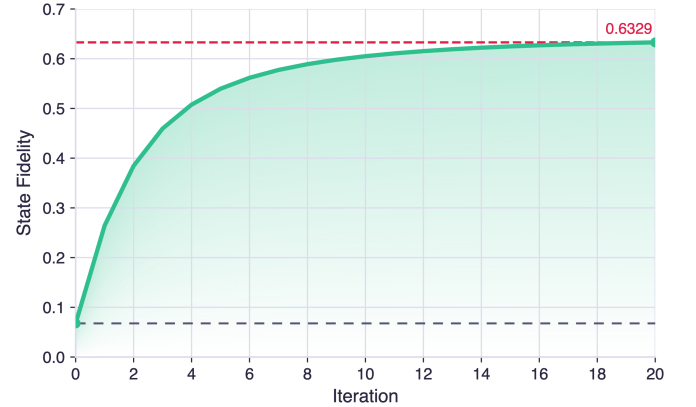


FIG. 4 FakeFез product-PTE convergence (8 qubits, 42 parameters). The trajectory descends over the first 5–10 iterations and flattens at the structural-well floor; dashed line marks $F = 0.6329$.

On the FakeFез simulator, the full TOPAZ LPTE ansatz (21 two-qubit gates, depth 97; Table VII) reaches ideal fidelity 0.9829 (approximation loss 1.71%) and, at its best seed, 0.9085 under noise (noise loss 7.41%; the 10-seed mean after pruning is 0.737 ± 0.066); exact preparation loses 14.99% under identical noise. These simulator values are for the 21-gate ansatz; the circuit submitted to live hardware is a distinct, more aggressively pruned instance (19 two-qubit gates, depth 76) whose noiseless one-way fidelity is 0.5662 (§6.6)—the two numbers describe different circuits, not conflicting reports of the same one. To make the relationship between them precise: the submitted

TABLE VII Comparison on FakeFez (quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion-test fidelity while using roughly half the gates. LPTE DM/Inv./CDR/VD are best-seed values of the trajectory in Fig. 4; the 10-seed mean after pruning is 0.737 ± 0.066 (Table I).

Method	DM	Inv.	CDR	VD	2Q	Depth
Exact prep (raw)	0.8501	0.7150	0.2313	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 lyr.)	0.1517	0.1305	—	—	28	57
TOPAZ product-PTE	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda = 0$, not measured. CDR and VD are single-method results applied to the LPTE circuit; they were not run stacked with ZNE. All mitigation methods (ZNE, CDR, VD) are implemented in-house and run against our own density-matrix simulations. The Exact-prep CDR entry (0.2313) should be read with care: CDR learns a linear model by regression on Clifford circuits sampled near the target, and at the depth-195 exact-preparation circuit that learned response leaves the linear-response regime CDR assumes, so its CDR estimate is unreliable; the shallow depth-97 LPTE circuit is precisely what keeps CDR inside its validity regime (0.9349).

instance is *not* a separately re-optimized seed. It is the converged endpoint of the same per-pair optimization that yields the simulator ansatz, and its parameter trajectory self-prunes within the run—the intensity weights for one entangling pair collapse to ρ -sum $\sim 4 \times 10^{-4}$ (deleted by the polar-projection gradient, §5.5.2) and two further pairs reduce to single dominant Pauli terms, leaving 19 of the 21 two-qubit gates after transpilation onto the heavy-hex path. No pairs are excised post hoc, and no independent 19-gate optimization was performed; the 42-parameter vector that produced the hardware result is the same dual-MMA output recorded in the per-run JSONs. The lower noiseless fidelity of the submitted instance is therefore a self-pruning consequence: pruning reduces depth at some cost to ideal representation, and here that cost is paid on the noisy objective, whereas the 0.9829 is the ideal (noise-free) evaluation of the full 21-gate ansatz before the depth-for-fidelity trade is made. LPTE is about twice as resilient to noise as the exact unitary (Fig. 4; product-PTE floor $F = 0.6329$). To quantify robustness across noise environments, parameters optimized under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The transfer efficiency

$$\eta = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{\bar{F}_{ij}}{\bar{F}_{ii}}, \quad (7)$$

is the mean ratio of off-diagonal (cross-model) fidelity $\bar{F}_{ij}$ to diagonal (on-model) fidelity $\bar{F}_{ii}$, where $\bar{F}_{ij}$ is the mean DM fidelity over 10 seeds of parameters trained under model i and evaluated under model j ; diagonal entries ($i = j$) are excluded to avoid self-counting. The result is $\eta = 0.9943$ (Fig. 5): parameters trained under one model retain 99.4% of their on-model fidelity under any other. The scope of this transfer claim is qualified: it is demonstrated within the same parametrized noise-model family (the six synthetic variants of

§4.4.2), not across categorically different noise channels such as correlated two-qubit faults or non-Markovian dynamics, whose cross-family transfer is untested here. Within that family, the structural well is not an overfit to any specific noise model.

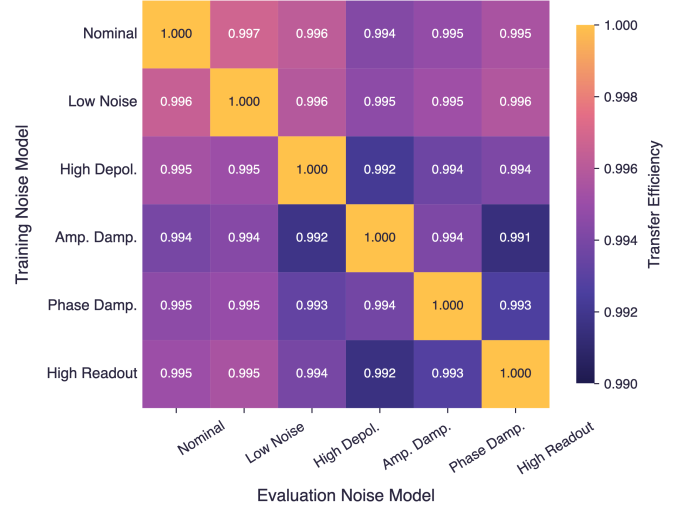


FIG. 5 Noise-model transfer matrix: DM fidelity of parameters trained under model i (rows) and evaluated under model j (columns), averaged over 10 seeds. Mean off-diagonal transfer efficiency $\eta = 0.9943$.

6.6 Real hardware on ibm_fez

Lemma 2 (Lipschitz equivalence gap). *This is a standard Lipschitz-type bound of the fidelity under a compiled-gate gap, stated here as a lemma for convenient reference rather than as a new result. The optimizer evaluates the polar-projected U_{safe} , while hardware executes the compiled Trotter product U_{circ} with $\|U_{\text{safe}} - U_{\text{circ}}\| \leq \epsilon$ [16]. By Fuchs–van de Graaf [37], $D \leq \sqrt{1 - F} \leq \sqrt{c} \epsilon$, so*

$$|L_{\text{ideal}} - L_{\text{hardware}}| \leq K \sqrt{c} \epsilon. \quad (8)$$

Because ϵ is small by design, Eq. (8) bounds how far the hardware landscape can drift from the simulator landscape.

The optimized (ρ, τ) from FakeFez are submitted to live `ibm_fez` (Heron r2, 156 qubits), where the experiment itself runs on the 8-qubit quietest path. State preservation is measured with a target-echo (inversion test) protocol: prepare the LPTE state with the optimized ansatz, then apply $U_{\text{target}}^\dagger$ and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of $U_{\text{target}}^\dagger$, so it is a strictly harder experiment than single-pass preparation. All hardware runs use `opt_level = 3` transpilation (the same level as all simulated results; see Table VIII) on path [140, 141, 142, 143, 144, 145, 146, 147], the quietest path at measurement time. The optimized ansatz uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline (the target itself compiles to 21 gates at depth 82) and 14 gates at depth 56 for

the MMA-optimized baseline. Two product-Trotter baselines are measured on the same day for a clean comparison: an *untuned* one with random gate parameters (the natural baseline, matching a naive product-Trotter submission) and an *optimized* one using the same dual-MMA routine TOPAZ applies to its own ansatz—isolating topology from parameter optimization. Both optimizations (LPTE and the MMA-optimized product baseline) ran the same dual-MMA solver on the same noise-aware objective with the same 30-iteration budget (recorded in Table VIII and the per-run JSONs), so the baselines are matched in optimizer and budget, not merely in name. The reader may ask how a circuit with the *same* topology as the target but fewer two-qubit gates (14 gates, depth 56, versus the target’s 21 gates at depth 82) can reach only 0.1768 noiseless fidelity—and, more pointedly, how it can reach *less* than the LPTE form (19 gates, depth 76, 0.5662) despite being shallower. The answer is not an expressibility ceiling: the product-Trotter ansatz is itself a product of the same $M = 21$ nearest-neighbour Pauli rotations used to define the target, so it can in principle represent it to arbitrarily high fidelity. The 0.1768 is instead the value the dual-MMA solver reaches on the *product-form* landscape under the matched budget—the same budget on which the per-pair (LPTE) form reaches 0.5662—so the gap is an optimization/landscape gap rather than a representational one. This is exactly the claim the paper advances: LPTE’s relaxed per-pair form is easier for the optimizer to search, and the hardware gain is a co-design consequence of topology and search, not an artifact of an under-converged baseline. The simulator product-PTE value 0.6329 (Table VII) is not in conflict with the hardware 0.1768: those are different circuits (simulator product-PTE is 8 gates at depth 22; the hardware product baseline is 14 gates at depth 56 after transpilation onto the heavy-hex device), evaluated under different measures (DM fidelity under FakeFey noise versus noiseless one-way fidelity), so a reader should not expect them to coincide.

TABLE VIII Live `ibm_fez` results, three same-day replications on the quietest path (2026-09-07, path [140, . . . , 147], `opt_level` = 3, $n = 2000$ shots per measurement). Measurement SEMs are binomial ($\sqrt{F(1-F)/n}$); the TOPAZ echo is reported as mean $\pm$ standard deviation across the three sessions (individual values in §6.6). Computed (noiseless) entries are exact statevector/density-matrix evaluations of the submitted circuit and carry no shot noise.

Configuration	Type	Fidelity
Product-Trotter, untuned random params (two-way echo)	measured	0.0478 $\pm$ 0.0034
Product-Trotter, MMA-optimised (two-way echo)	measured	0.1008 $\pm$ 0.0048
TOPAZ LPTE (two-way echo, mean of 3 sessions)	measured	0.2883 $\pm$ 0.0394
TOPAZ LPTE (two-way echo, individual sessions)	measured	0.2443 $\pm$ 0.0068, 0.3008 $\pm$ 0.0073, 0.3200 $\pm$ 0.0074
ZNE on TOPAZ echo (per session)	measured	0.2058, 0.3075, 0.3968
TPNM on TOPAZ echo (per session)	measured	0.2919, 0.3159, 0.3200
TOPAZ LPTE (one-way)	noiseless	0.5662
Product-Trotter, MMA-optimised (one-way)	noiseless	0.1768
Product-Trotter, untuned random (one-way)	noiseless	0.0758

To disambiguate the quantities reported here: the top seven rows are device measurements ($n = 2000$ shots each). $F_{\text{echo}} = 0.2883 \pm 0.0394$ is the mean measured target-echo—the return probability after $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$ on device, a protocol that doubles the forward gate count plus the transpilation overhead of $U_{\text{target}}^\dagger$. The three sessions (0.2443, 0.3008, 0.3200) span a sin-

gle overnight window on the same quietest path, and their spread is the same order as the binomial SEM—the measurement is stable across replications. The one-way entries are not device measurements but the fidelity of the *hardware-submitted circuit* (19 two-qubit gates, depth 76) against the ideal target, evaluated exactly without shot noise: $F_{1\text{-way}} = 0.5662$ is the noiseless one-way state-preparation fidelity $|\langle \psi_{\text{target}} | \psi_{\text{circ}} \rangle|^2$. Because these are computed quantities they are reported without error bars. They are not interchangeable with the echo: the echo is a lower bound on the noisy one-way fidelity, not equal to it. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along the path.

6.6.1 The inversion-test echo and the role of ZNE

The hardware experiment measures a *target-echo* (inversion test): the return probability of $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$, which quantifies how closely the optimized state matches the target. Unlike a self-echo $U \cdot U^\dagger$ (which is trivially 1 for any unitary), the target-echo measures state-preparation fidelity directly through the round-trip protocol, so $F_{\text{echo}} = 0.2883 \pm 0.0394$ is a lower bound on the noisy one-way fidelity ($F_{1\text{-way}} = 0.5662$). The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along the path.

Relative to the same-day, same-path untuned product-Trotter baseline (0.0478 ± 0.0034), the mean TOPAZ raw echo is a $6.04\times$ gain ($z \approx 22$, $p < 10^{-7}$; Fig. 6); against the MMA-optimised product baseline (0.1008 ± 0.0048) it is a $2.86\times$ gain ($z \approx 17$). Neither comparison is confounded by unequal representational capability: against the *same* target, the untuned product baseline’s noiseless one-way fidelity is 0.0758 and the optimised baseline’s is 0.1768, so TOPAZ represents the target $7.47\times$ better than the untuned random-parameter circuit and $3.20\times$ better than the optimised one under noise-free evaluation (0.5662 vs. $0.0758/0.1768$). The raw-echo gain therefore tracks the noiseless representation gain, which the retention fractions confirm: under free evaluation both circuits retain comparable shares of their ideal fidelity through the noisy round-trip (50.9% for TOPAZ vs. 63.1% and 57.0% for the two baselines), so the $6.04\times$ raw-echo improvement is attributable to representation—not to any hidden advantage of a shallower circuit, since all three circuits are compared against the same inversion protocol.

Standard ZNE is *inconsistent* on this protocol: across the three sessions the extrapolated echo ranges from below the raw echo (0.2058 vs. 0.2443) to above it (0.3968 vs. 0.3200), and the mean never approaches the noiseless bound. Tensor-product noise-model (TPNM) correction—a noise-aware estimate that models the round-trip channel as a tensor product of per-gate depolarizing channels—behaves more stably (0.2919, 0.3159, 0.3200), tracking the raw echo across sessions. TOPAZ delivers its gain before noise accumulates, on the circuit itself, so the pre-circuit optimization is complementary to—and, on this protocol, more reliable than—extrapolation-based post-processing.

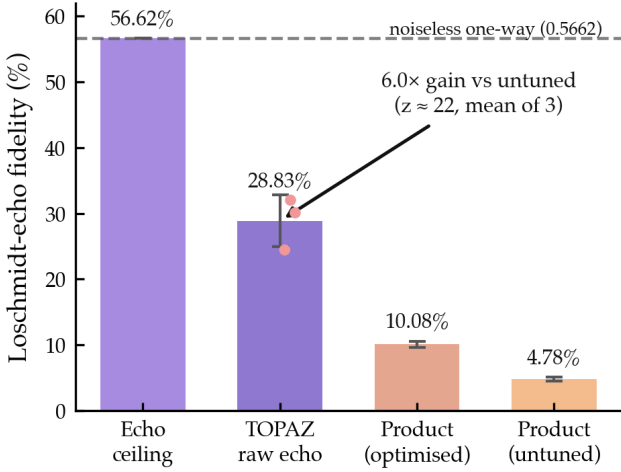


FIG. 6 Live `ibm_fez` target-echo (inversion test): TOPAZ raw echo versus product-Trotter baseline and standard ZNE.

6.7 Stacking with post-processing QEM

TOPAZ composes with post-processing because the two act on different parts of the pipeline: TOPAZ pre-tunes the parameter space, while ZNE, CDR, and VD correct observables. ZNE alone at the baseline gives marginal improvement ($0.0677 \rightarrow 0.0747$); on top of TOPAZ it adds a further 0.0667 DM gain. The operating point moved by TOPAZ lies on a part of the noise curve where extrapolation is more reliable. As in Table VII, ZNE, CDR, and VD are each applied to the LPTE circuit individually (our own implementations, run against the same simulated noise model); we did not run them as a combined stack, so the Table IX rows are single methods rather than a full mitigation stack.

TABLE IX 8-qubit mitigation stacking on FakeFaz (LPTE baseline). DM: density-matrix fidelity under the target noise model; Inv.: inversion-test fidelity (target-echo).

Configuration	DM Fidelity	Echo Fidelity
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9085	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

Each mitigation is applied individually to the LPTE circuit; a combined ZNE+CDR+VD stack was not run, so the rows are not additive. “TOPAZ LPTE alone” is the same best-seed 21-gate circuit as Table VII (DM 0.9085, echo 0.7715).

6.8 Scaling to 16 qubits

To probe the representability ceiling of the LPTE ansatz beyond 8 qubits, we ran the full TOPAZ pipeline on `ibm_fez` (Heron r2) for a 16-qubit nearest-neighbour chain [112, 113, 131, 132, 133, 114, 115, 129, 130, 117, 116, 145, 146, 109, 110, 111] using the *per-pair* mode (90 parameters). The optimizer improves

the return probability from 0.0003 (random) to 0.1151—a $383\times$ gain from the classical search alone—but this noiseless ceiling (11.5%) is far below the 56.6% achieved on the 8-qubit path, confirming that the LPTE ansatz’s expressibility collapses as the chain lengthens and the linear topology becomes increasingly incompatible with the heavy-hex device connectivity. Read through the finite-depth characterization of §2, this is a *structural boundary* of the linear-chain form: a 16-qubit chain adds nine blocks to the seven of the 8-qubit case—more than twice the $\mathfrak{su}(2)$ copies of Eq. (1)—but the added blocks sit on a longer path whose gate-depth cost outgrows the per-block entanglement gain at fixed device coherence, the same depth-versus-coherence trade that the 8-qubit experiments resolve in TOPAZ’s favour at shallower depth.

The unmitigated target-echo (raw echo) is 0.0065; TPNM correction gives 0.0103; our in-house ZNE (standard unitary folding, $SF = 1, 3$, single batched job) gives 0.0120. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.0065/0.1151 = 5.6\%$, well below the $\sim 51\%$ observed at 8 qubits. A separate ZNE-only run using the standard IBM unitary-folding protocol ($SF = 1, 3, 5$, Richardson extrapolation) yielded folds $0.00475/0.00625/0.0030$ —non-monotonic, the same inconsistency signature seen at 8 qubits (where the extrapolated echo ranged from below to above the raw echo across sessions). The Richardson fit extrapolated to 0.001497, which when clipped to the noiseless bound gives 0.002219, collapsing below the raw echo. ZNE’s instability on these deep round-trip circuits is therefore structural (intrinsic to deep round-trip circuits) rather than an implementation artifact: gate stretching pushes the circuit beyond the coherence window, and extrapolation frequently produces a result at or below the unmitigated measurement.

These 16-qubit results serve as a scaling-limitation data point: the LPTE ansatz’s representability ceiling (11.5% noiseless fidelity) and the ZNE failure mode both worsen with qubit count, consistent with the deep-circuit decoherence picture. Extending the framework to larger systems would require adaptive pruning thresholds and device-side optimization—directions we discuss in §8.

6.9 Scaling measurements: gradient variance and the closed algebra

To separate trainability from the noise-limited experiments above, we measured two noiseless, statevector scalings directly, without the TOPAZ pipeline, on random instances of the three parametrizations. Both are pure measurements; no asymptotic classification is attached to either. The first is the variance (over parameters at fixed random initialization) of the first-order central finite-difference of the state-preparation fidelity $F = |\langle \psi_{\text{target}} | \psi \rangle|^2$, at step 10^{-6} , for LPTE chains, product-PTE chains of identical pair structure, and a hardware-efficient ansatz (HEA) of RY+CNOT layers on the same linear chain. Ten random initializations are used at each $N \in \{4, 6, \dots, 16\}$, with $N - 1$ nearest-neighbour blocks (six parameters per LPTE/product pair, so the LPTE and product curves share an exact parameter count; the HEA carries its native two single-qubit rotations per layer and so a nearby,

not identical, count). Figure 7 shows the result: on a $\log_2$ scale the decay is linear in N for all three, with the LPTE chain decaying most slowly. Least-squares slopes (95% CI) are -1.5466 ± 0.0791 (LPTE), -2.2905 ± 0.2000 (product), -1.9627 ± 0.1078 (HEA), all $R^2 \geq 0.99$. At $N = 16$ the $\log_2$ variances are -21.75 (LPTE), -36.08 (product), -33.53 (HEA)—an approximately $2.0 \times 10^4 \times$ gap between LPTE and product PTE, and $3.5 \times 10^3 \times$ over the HEA, in raw variance of the fidelity gradient. Equivalently, each added qubit attenuates the LPTE gradient variance by a factor $2^{-1.55} \approx 0.34$, versus $2^{-2.29} \approx 0.20$ for product PTE and $2^{-1.96} \approx 0.26$ for the HEA. The same ordering (LPTE slowest decay) holds pointwise at every N except a single crossover between the product and HEA curves at $N = 10$. We report these numbers as measurements of the three parametrizations on this target family; they do not by themselves establish escape from barren-plateau behaviour, which additionally requires the cost function and parameter-ansatz conditions of Refs. [17, 18].

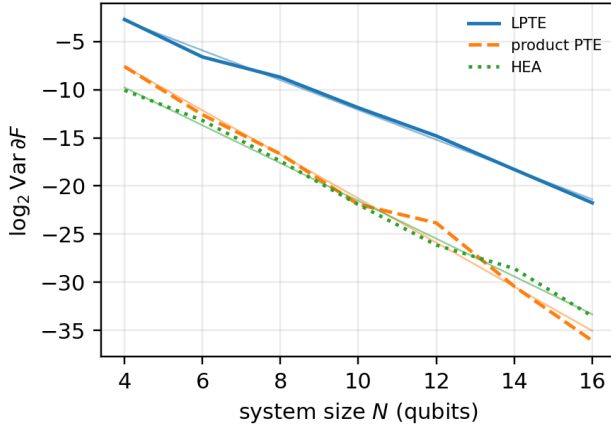


FIG. 7 Noiseless scaling of fidelity-gradient variance versus system size for LPTE (solid), product PTE (dashed), and a hardware-efficient ansatz (dotted), $\log_2$ scale, ten random initializations per point. Lines are least-squares fits: slopes -1.55 (LPTE), -2.29 (product), -1.96 (HEA).

The second measurement completes the finite-depth characterization of §2: the dimension of the Lie closure generated by the LPTE tangent-space generators at the identity, i.e. the Dynamical Lie Algebra (DLA) of the chained ansatz. The generators are the per-block $\{XX, YY, ZZ\}$ triples of the $N - 1$ nearest-neighbour blocks, which are vectors of Pauli strings closed under commutation. We computed the exact Lie closure by breadth-first commutator enumeration over Pauli-word bases—avalanched vectorized over all 4^{2N} strings for $N \leq 8$, giving exact dimensions—and cross-checked each closed result against the exact matrix-based Lie span for small N . Table X reports the exact dimensions.

Across $N = 4$ through 8 the closure dimension follows $\dim \mathfrak{l} = 4^{N-1} - 1$ for odd N and $4^{N-1} - 4$ for even N —for large N these are $[4^N - 1]/4 - 3/4$ and $[4^N - 1]/4 - 15/4$, i.e. one quarter of the full $\mathfrak{su}(2^N)$ dimension up to a parity-dependent

TABLE X Measured dimension $\dim \mathfrak{l}$ of the Lie closure of the $3(N - 1)$ per-pair generators of the LPTE chain, compared with the maximum $\dim \mathfrak{su}(2^N) = 4^N - 1$. Rows $N \leq 8$ are exact (enumeration completed).

N	$\dim \mathfrak{l}$	$\dim \mathfrak{su}(2^N)$	$\dim \mathfrak{l} / \dim \mathfrak{su}$
4	60	255	0.2353
5	255	1023	0.2493
6	1020	4095	0.2491
7	4095	16383	0.2499
8	16380	65535	0.2499

constant term. The fraction $\dim \mathfrak{l} / \dim \mathfrak{su}$ is 0.235 at $N = 4$ and converges toward 0.2499 by $N = 8$. For $N = 9$ to 12 the commutator budget capped the enumeration, yielding lower bounds (e.g. 65475 at $N = 9$, 0.250 of the ambient dimension) that are monotone increasing but no longer exact; we report them as bounds only. This is a measured characterization, not an asymptotic theorem: it states what the chained $\{XX, YY, ZZ\}$ block algebra actually closes to at the finite depths used here, which is what the per-block $\mathfrak{su}(2)$ picture of §2 and the two-band canyon-river geometry (§7.1) are consistency checks of. One quarter of the full algebra is still exponentially large in N , so the measurement is neutral on the question of whether the block-restricted growth delays saturation of the full algebra; consistent with §2, we attach no barren-plateau conclusion to it.

7. Well geometry and sparsity

Across every seed, session, and backend (triple-channel, FakeFez, FakeMontrealV2, live ibm_fez), the trajectory shows the same shape: a rapid descent over the first 5–10 iterations, then a stable floor. The well is not a three-dimensional attractor but a family of sparse subspaces: each seed prunes to a different 8–24 dimensional subset of the 42-dimensional (ρ, τ) space, and the survivors are not the target Hamiltonian’s largest coefficients—the top six target terms are all pruned. The optimizer selects for noise robustness, not target matching. Figure 8 makes the family-of-wells picture concrete: no two seeds prune the same set of pairs.

7.1 Canyon-river curvature

We use the term *canyon-river* to describe the heterogeneity pattern in the Hessian eigenvalues at the well: the intensity subspace is uniformly steep ($\kappa_\rho \approx 1$, all ρ equally stiff at ~ 870), while the timing subspace has a few extremely stiff directions amid many soft ones ($\kappa_\tau \approx 9 \times 10^3$, timings vary by $8,000 \times$ among survivors), with combined $\kappa \approx 2 \times 10^5$. The eigenvalue clustering into two sharp bands is consistent with dampened barren-plateau behaviour [17], but no eigenvalue-versus-qubit-

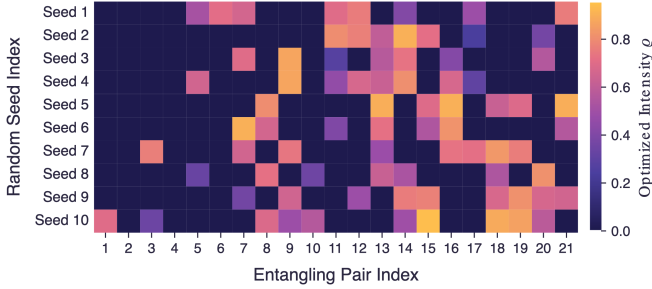


FIG. 8 Family-of-wells sparsity: final pair intensities $\kappa(p) = \sum_P r_{p,P}$ for 10 random seeds (rows) across the 21 LPTE entangling pairs (columns). Dark cells are pruned parameters ($\rho < 0.05$); each seed retains a different sparse subset.

count diagnostic was performed, so we make no claim of barren-plateau avoidance [18]. The two-band geometry is also the shape expected from the finite-depth per-block $\text{su}(2)$ picture of §2: a few stiff directions inherited from the actively optimized blocks riding atop many soft ones.

8. Discussion

8.1 Scope and limitations

All headline circuits are 8 qubits on a linear chain—a deliberate first demonstration: the quietest path of the Heron r2 device (path [140, ..., 147]) is itself a nearest-neighbour chain, so the experiment matches the device’s natural locality, and linear chains are the unit cells out of which heavy-hex topologies are assembled. The sparsification means are over 5 seeds (10 for the ablation). The pruning threshold $\tau_\kappa = 0.05$ is a heuristic choice; a quantitative sensitivity study of τ_κ (gate reduction and fidelity versus threshold) is planned but not yet reported, and systematic threshold sweeps across molecule sizes and noise levels remain future work. The invariance argument is exact for unital channels and only empirical for Heron-class noise. The 8-qubit headline experiment was conducted within a single overnight window on 2026-09-07 (three same-day replications plus both baselines on the same quietest path); the preliminary 16-qubit experiment was conducted on a separate day (2026-09-04). Reproducibility across different calibration sessions and hardware revisions is not yet established. A preliminary 16-qubit run (§6.6.8) confirmed the representability ceiling (11.5% noiseless fidelity vs. 56.6% at 8 qubits) and the same standard-ZNE instability (extrapolation collapsing below the raw echo), indicating that the 8-qubit headline results do not straightforwardly extend to longer chains without adaptive pruning and device-side optimization. Global PTE is computationally expensive for $N \geq 4$ and was not used for the headline experiments; a from-scratch TOPAZ run on critical-point TFIM still leaves a 0.4–1.2 Ha residual. Transfer across physical layouts (e.g. from a linear chain to a heavy-hex topology) was not explored; the current demonstration tests noise-model transfer only. The density-matrix engine makes simulation-path scaling intractable beyond ~ 20 qubits; the transfer efficiency $\eta = 0.9943$ mitigates this by letting offline-

trained parameters move to hardware without re-optimization. The open problem is a device-side MMA iterate for cases where the simulator diverges from hardware noise at scale.

8.2 Relation to post-processing QEM

ZNE, CDR, and virtual distillation act on measurement statistics after the circuit runs; TOPAZ acts on the topology before execution. The two are complementary rather than competitive. Empirically, TOPAZ circuits preserve the noise-energy relationship that extrapolation methods rely on, so stacking TOPAZ with a post-processor improves that method’s reliability rather than merely adding a constant.

8.3 Relation to adaptive circuit-topology methods

The idea that topology itself should be optimized—rather than fixed at ansatz construction—has several near-term antecedents, summarized in Table XI. ADAPT-VQE [23] and qubit-ADAPT-VQE [38] *grow* an ansatz by greedily appending the highest-gradient operator from a pool, with no pruning stage. Pruned-ADAPT-VQE [39] adds a discrete, post-hoc removal of near-zero-amplitude operators from a grown ansatz—still inside the discrete operator-pool paradigm, with no continuous relaxation and no noise-awareness. Plateau-aware operator elimination [40] removes non-contributing pool operators once detected, again discretely. TopGen [41] generates circuit topologies bottom-up by combining sub-circuits selected for accuracy, optimizing the topology through a discrete search over a circuit library. TOPAZ differs on three axes. (i) *Continuity*: edge density is a real-valued design variable, so gradients flow during edge removal; the adaptive-VQE family and TopGen prune or select discretely. (ii) *Granularity and timing*: pruning operates on the entangling graph at the level of per-pair intensity *before* compilation, so the exported circuit never carries idle edges, whereas ADAPT-style pruning removes operators from an already-grown pool. (iii) *Objective*: TOPAZ is trained against a noise-aware density-matrix objective, while the adaptive-VQE family optimizes ideal energy and TopGen optimizes ideal accuracy. Table XI makes the contrast explicit.

TABLE XI Positioning TOPAZ against prior approaches that adapt circuit structure. “Continuous”: edge presence is relaxed to a real-valued intensity during optimization. “Prune stage”: unused operators/edges are removed at some point in the pipeline.

Method	Selection	Continuous	Prune stage	Noise-aware
ADAPT-VQE [23]	grow, max-gradient pool	no	no	no
Qubit-ADAPT-VQE [38]	grow, max-gradient Pauli pool	no	no	no
Pruned-ADAPT-VQE [39]	grow + amplitude removal	no	yes (post-hoc)	no
Plateau-operator elimination [40]	grow + operator removal	no	yes (in-pool)	no
TopGen [41]	discrete bottom-up search	no	yes (sub-circuit merge)	no
TOPAZ (this work)	continuous density + polar projection	yes	yes (pre-compile, τ_κ)	yes

8.4 Future work

Several directions extend the current framework. (i) Device-side optimization: replacing the simulator-based dual-MMA with a hardware-in-the-loop iterate would eliminate the simulator-hardware gap and enable scaling beyond the ~ 20 -qubit simulation ceiling. (ii) Adaptive pruning: replacing the

fixed threshold $\tau_k = 0.05$ with a schedule that adapts to the loss landscape curvature could improve gate reduction on larger instances. (iii) Non-linear topologies: extending LPTE beyond linear chains to heavy-hex or modular architectures would match the connectivity of production hardware. (iv) Larger molecules: applying TOPAZ to systems beyond H_2/H_4 —such as LiH , BeH_2 , or small active-space problems—would test whether the structural-well phenomenon persists at greater chemical complexity. (v) Combined mitigation stacks: running ZNE+CDR+VD simultaneously (rather than individually) on TOPAZ circuits would quantify whether the methods’ gains are additive, sub-additive, or synergistic.

9. Conclusion

We have presented TOPAZ, a classical framework that optimizes quantum circuit topology by treating entangling-edge intensities as continuous design variables. Its central structural result is the topological-sparsification theorem (Theorem 1): an idle entangling edge is deleted by the polar projection as a continuous, exact consequence of the relaxation, so pruning is not a discrete heuristic. The optimizer converges to structural wells—formally the sublevel-set basins of Definition 1, measured at finite depth through the chained algebra’s closure to one quarter of $\mathfrak{su}(2^N)$ (§6.6.9). On the 8-qubit FakeFey simulator, TOPAZ’s pruning achieves a 34% depth reduction relative to its own unpruned ansatz at negligible fidelity cost. The durable and reproducible advantage, isolated by the matched-budget re-evaluation of §6.6.4, is *mitigation-compatibility*: ZNE/CDR remain physically valid on the shallow topology without clamping, whereas the deeper hardware-efficient form required clamping on the majority of seeds at a matched budget. On the 156-qubit *ibm_fez* processor, a scoped 8-qubit illustration on one quietest path attains a 2.9× improvement in target-echo over the MMA-optimized product-Trotter baseline under matched parameter optimization, and 6.0× over the untuned baseline (mean of three same-day sessions, $z \approx 22$); against the same target, the shallowest submitted circuit—the 14-gate optimized product baseline—realises 3.2× less of the ideal target fidelity than TOPAZ’s 19-gate circuit (0.1768 vs. 0.5662 noiseless), so the hardware gain is conditional on topology selection, not on depth reduction alone, with a 50.9% retention fraction of the one-way fidelity. The structural well transfers across synthetic noise models ($\eta = 0.9943$). A preliminary 16-qubit experiment confirmed both the representability ceiling (11.5% noiseless fidelity, down from 56.6% at 8 qubits) and the same standard-ZNE instability, framing the 8-qubit experiment as a proof of concept whose extension to larger systems remains future work. The framework is limited to 8-qubit linear chains and hardware demonstrations spanning two sessions (2026-09-04 for the 16-qubit probe and 2026-09-07 for the 8-qubit illustration); nevertheless, the results establish that circuit topology—not just gate parameters—is a design variable worth optimizing.

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Author contributions

K.K. conceived the study, developed the TOPAZ framework and the parametric Trotter expression family, designed and ran the simulations and hardware experiments, and wrote the manuscript.

Competing interests

The author declares no competing interests. The author holds a junior-researcher position at Bloq Quantum; this affiliation played no role in the design, execution, or reporting of the work.

Data availability

All code, data, and analysis scripts for this work are available at the project’s public repositories (TOPAZ [20]; the companion continuous-topology manuscript [19]; the companion PTE-UPTE derivation [12]). The TOPAZ release used for the results of this manuscript is tagged v1.0, archived with its exact commit hash and full dependency environment in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449). Qiskit ≥ 0.46 and Qiskit Aer ≥ 0.14 were used; the pinned versions are recorded in `environment.yml`. Table-to-script mappings: Table V data is in `python/0-FINAL/results_h2_vqe/` and `python/0-FINAL/results_h4_vqe/` (per-seed logs in `FINAL_PYTHON/rerun_logs/`); Table VI is produced by `experiments/h4_hear_fair_noisy_eval.py`, per-seed record in `experiments/results/h4_hear_fair_noisy_eval.json` (“noiseless”= $(E_{\text{vqe}} - E_{\text{FCI}})$ at the converged parameters, “noisy” at $n=2000$ density-matrix shots, “ZNE+CDR” as in Table V); Table VII and Table II are produced by `benchmark_lppe.py`; Table I by `ablation.py`; Table VIII by `run_ibmfez.py`; the 16-qubit data in `results_16q_hw/`; Figs. 4, 1, 2, 5, 6, 8 by the corresponding `plot_*` scripts in the same tree.

Hardware job records (Qiskit Runtime job IDs and per-circuit shot counts for each row of Table VIII, together with the backend calibration timestamps captured at submission) are stored in `data/ibmfez_runs.jsonl`. These records, including the unredacted job IDs and the exact v1.0 release commit hash, are archived in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449) at publication time, so that every circuit measurement in Table VIII is reproducible from the pinned code revision and raw runtime records.

Code availability

The optimization engine, Kraus contraction routines, and experiment drivers are released as open-source Python under the

TOPAZ repository (DOI 10.5281/zenodo.19214449, tag v1.0).

Additional information

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